

Week – 1 Finding Time complexity of algorithm

1.

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
```

```
{  
    int i= 1;  
  
    int s =1;  
  
    while(s <= n)  
    {  
        i++;  
        s += i;  
    }  
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

C program:

```
1 #include <stdio.h>
2
3 int count = 0;
4 void fun(int n)
5 {
6     int i = 1;
7     int s = 1;
8     count+=2;
9
10    while(s<=n)
11    {
12        i++;
13        count++;
14        s+=i;
15        count++;
16        count++;
17    }
18
19    count++;
20    printf("%d",count);
21
22 }
23
24
25 int main()
26 {
27     int n;
28     scanf("%d",&n);
29     fun(n);
30     //printf("%d",count);
31 }
```

	Input	Expected	Got	
✓	9	12	12	✓
✓	4	9	9	✓

Passed all tests! ✓

Correct

2.

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
        printf("*");
    }
    else
    {
        for(int i=1; i<=n; i++)
        {
            for(int j=1; j<=n; j++)
            {
                printf("*");
                printf("*");
                break;
            }
        }
    }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

C Program :

```
1 #include <stdio.h>
2 int count = 0;
3
4 void func(int n)
5 {
6     if(n==1)
7     {
8         count++;
9     }
10
11     else
12     {
13         count++;
14         for(int i = 1;i<=n;i++)
15         {
16             count++;
17             for(int j = 1;j<=n;j++)
18             {
19                 count+=3;
20                 break;
21             }
22             count++;
23         }
24         count++;
25     }
26 }
27
28
29 int main()
30 {
31     int n;
32     scanf("%d",&n);
33     func(n);
34     printf("%d",count);
35 }
```

	Input	Expected	Got	
✓	2	12	12	✓
✓	1000	5002	5002	✓
✓	143	717	717	✓

Passed all tests! ✓

3.

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {  
  {  
    for (i = 1; i <= num; ++i)  
    {  
      if (num % i == 0)  
      {  
        printf("%d ", i);  
      }  
    }  
  }  
}
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

C Program:

Answer:

```
1 #include <stdio.h>
2 int count = 0;
3
4 void fact(int n)
5 {
6     for(int i = 1; i <= n; i++)
7     {
8         count++;
9         if(n%i == 0)
10        {
11            count++;
12        }
13        count++;
14    }
15    count++;
16    printf("%d", count);
17 }
18
19
20 int main()
21 {
22     int n;
23     scanf("%d", &n);
24     fact(n);
25 }
```

	Input	Expected	Got	
✓	12	31	31	✓
✓	25	54	54	✓
✓	4	12	12	✓

Passed all tests! ✓

Correct

4.

Convert the following algorithm into a program and find its time complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

C Program:

```
1 #include <stdio.h>
2 int count = 0;
3
4 void fun(int n)
5 {
6     int c = 0;
7     count++;
8
9     for(int i = n/2; i < n; i++)
10     {
11         count++;
12         for(int j = 1; j < n; j = 2 * j)
13         {
14             count++;
15             for(int k = 1; k < n; k = k * 2)
16             {
17                 count++;
18                 c++;
19                 count++;
20             }
21             count++;
22         }
23         count++;
24     }
25     count++;
26     printf("%d", count);
27 }
28
29 int main()
30 {
31     int n;
32     scanf("%d", &n);
33     fun(n);
34 }
```

	Input	Expected	Got	
✓	4	30	30	✓
✓	10	212	212	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

5.

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
    int rev = 0, remainder;
    while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n /= 10;
    }
    print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

C program:

Answer:

```
1 #include <stdio.h>
2 int count = 0;
3
4 void rev(int n)
5 {
6     int rev = 0, rem;
7     count++;
8
9     while(n!=0)
10 {
11     rem = n%10;
12     rev = rev * 10 + rem;
13     n/=10;
14     count+=4;
15 }
16
17 count+=2;
18 printf("%d", count);
19 }
20
21 int main()
22 {
23     int n;
24     scanf("%d", &n);
25     rev(n);
26 }
```

	Input	Expected	Got	
✓	12	11	11	✓
✓	1234	19	19	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.